Task 1



Read the article from *The Week* magazine about 'A step closer to cloning babies' and answer the questions below. You will need to research each answer. There are some suggested links to help.

1) What are stem cells?
http://www.bionetonline.org/English/Content/sc_cont1.htm

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2) Where are stem cells found?

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3) What is an embryonic stem cell and why are they so useful in therapeutic treatments? http://learn.genetics.utah.edu/content/tech/stemcells/quickref/

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4) What is a blastocyst? Watch this presentation:
http://learn.genetics.utah.edu/content/tech/stemcells/scintro/

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5) What is Parkinson's disease and what are its symptoms?
http://www.nhs.uk/Conditions/Parkinsons-disease/Pages/Introduction.aspx

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6) What do you think stem cells could be used for in the future?

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7) Why do some people object to the use of stem cells?

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Task 2

Put the following statements into the correct order to describe the process of adult cell cloning. Add diagrams and annotations to the statements.

Give the cell a tiny electric shock to make it start dividing to form embryo cells.	The nucleus is removed from an unfertilised egg.
Insert the nucleus into the empty unfertilised egg cell.	The embryo develops into a ball of cells.
Take a nucleus from an adult body cell, e.g. a skin cell of the animal that you wish to clone.	The embryo is inserted it into the uterus of an adult 'host' female to continue its development.

Teaching notes

The webpage links were all available at the time of publication.
The accompanying article from p.23 of *The Week* magazine (issue 921, 25 May 2013) can be downloaded from www.teachitscience.co.uk.

Task 1 - answers

1. Stem cells are undifferentiated cells which can become any type of differentiated cell.

2. Stem cells are found in bone marrow and embryos.

3. A stem cell with the potential to form a number of different specialised cells, which is taken from an early embryo.

4. An embryo at a very early stage of development.

5. Parkinson's disease is a degenerative neurological condition. The symptoms are involuntary shaking, muscle stiffness and slowed physical movements.

6. Answers will vary but may include use of stem cells to treat diseases such as heart disease, neurological disorders, blood disorders and diabetes.

7. Object to using embryos which are potential human life. Answers that just say 'unethical' or 'for religious reasons', need more explanation.

Task 2 - answers

The nucleus is removed from an unfertilised egg.
Take a nucleus from an adult body cell, e.g. a skin cell of the animal that you wish to clone.
Insert the nucleus into the empty unfertilised egg cell.
Give the cell a tiny electric shock to make it start dividing and forming embryo cells.
The embryo develops into a ball of cells.
Insert it into the uterus of an adult 'host' female to continue its development.